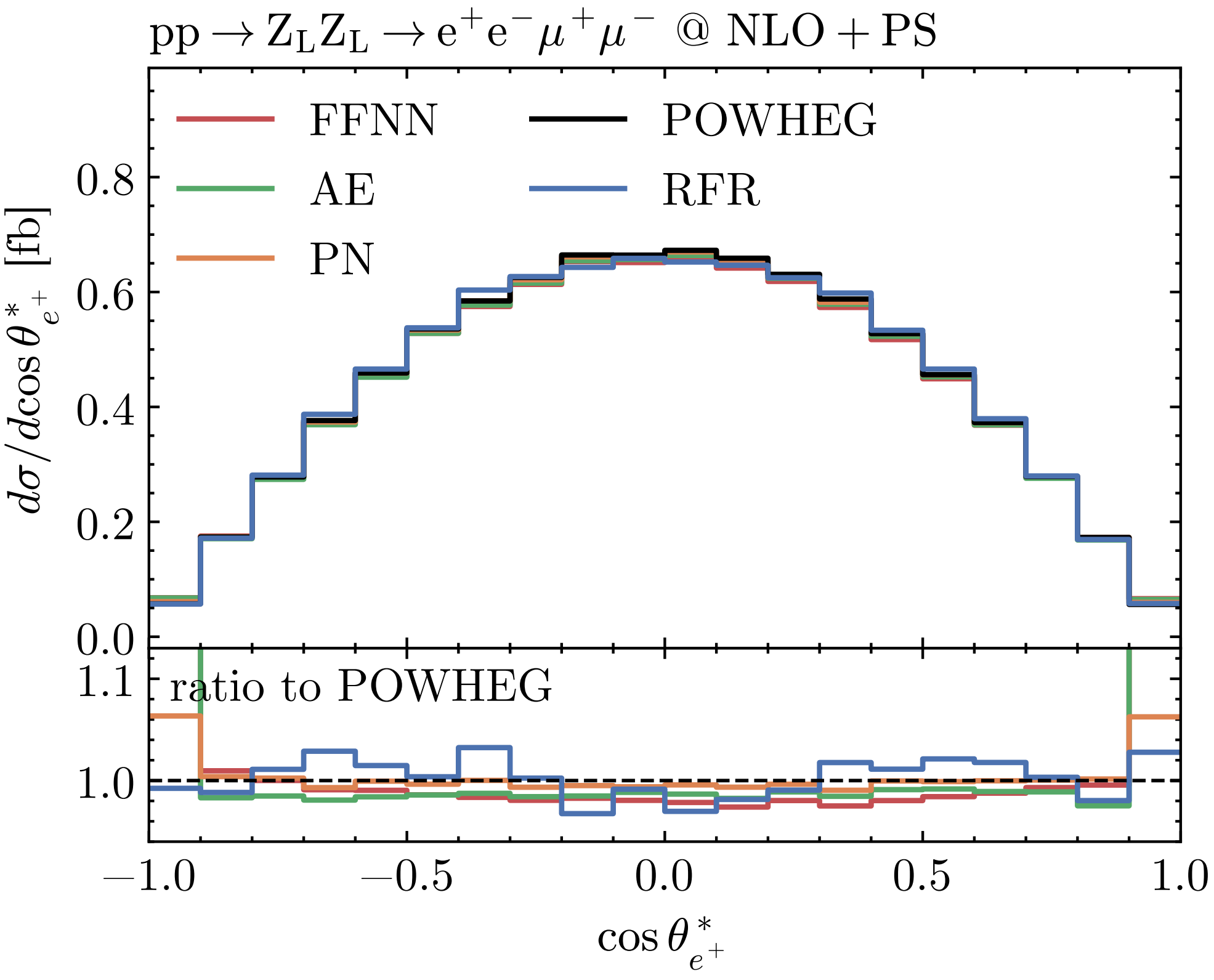
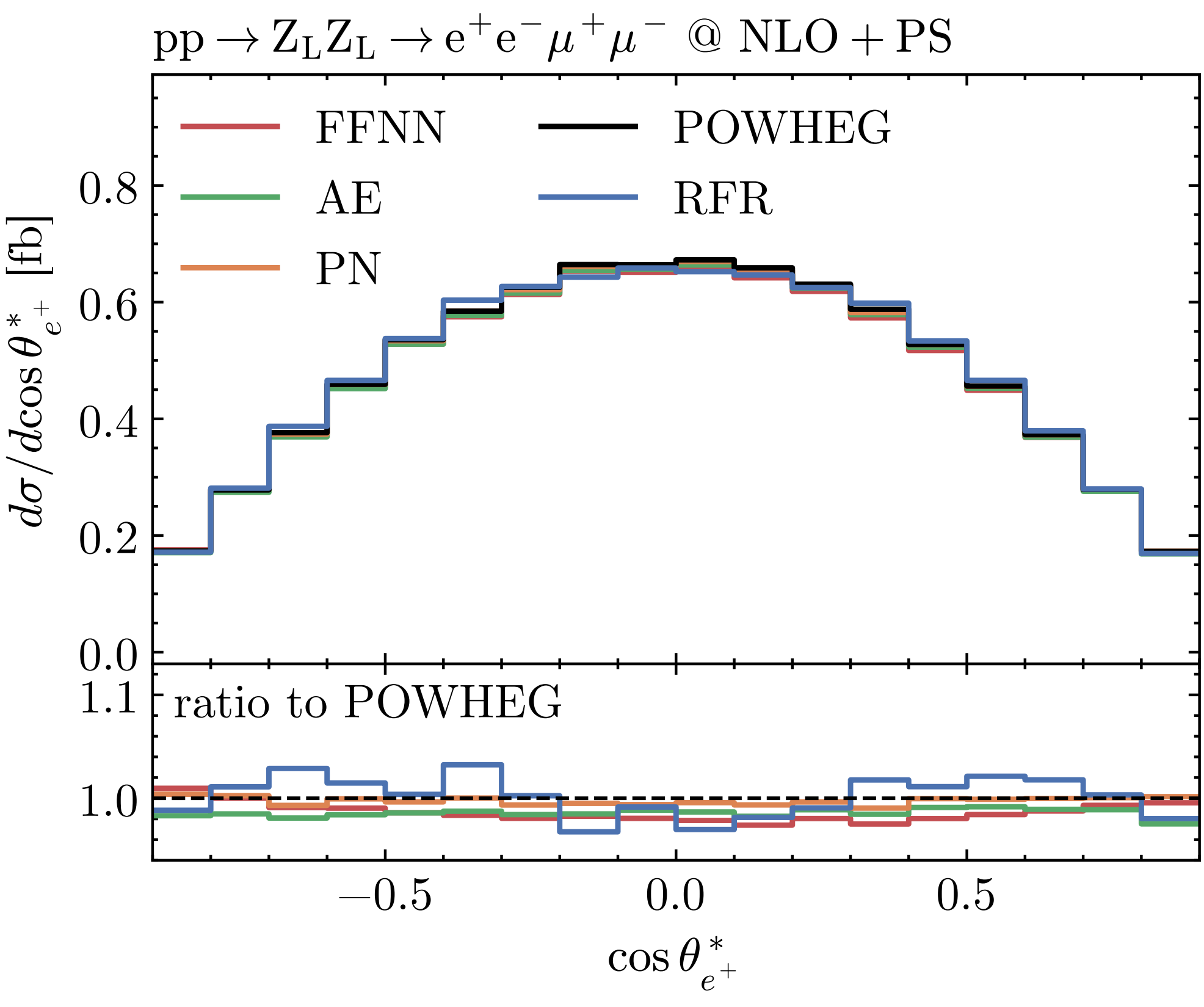
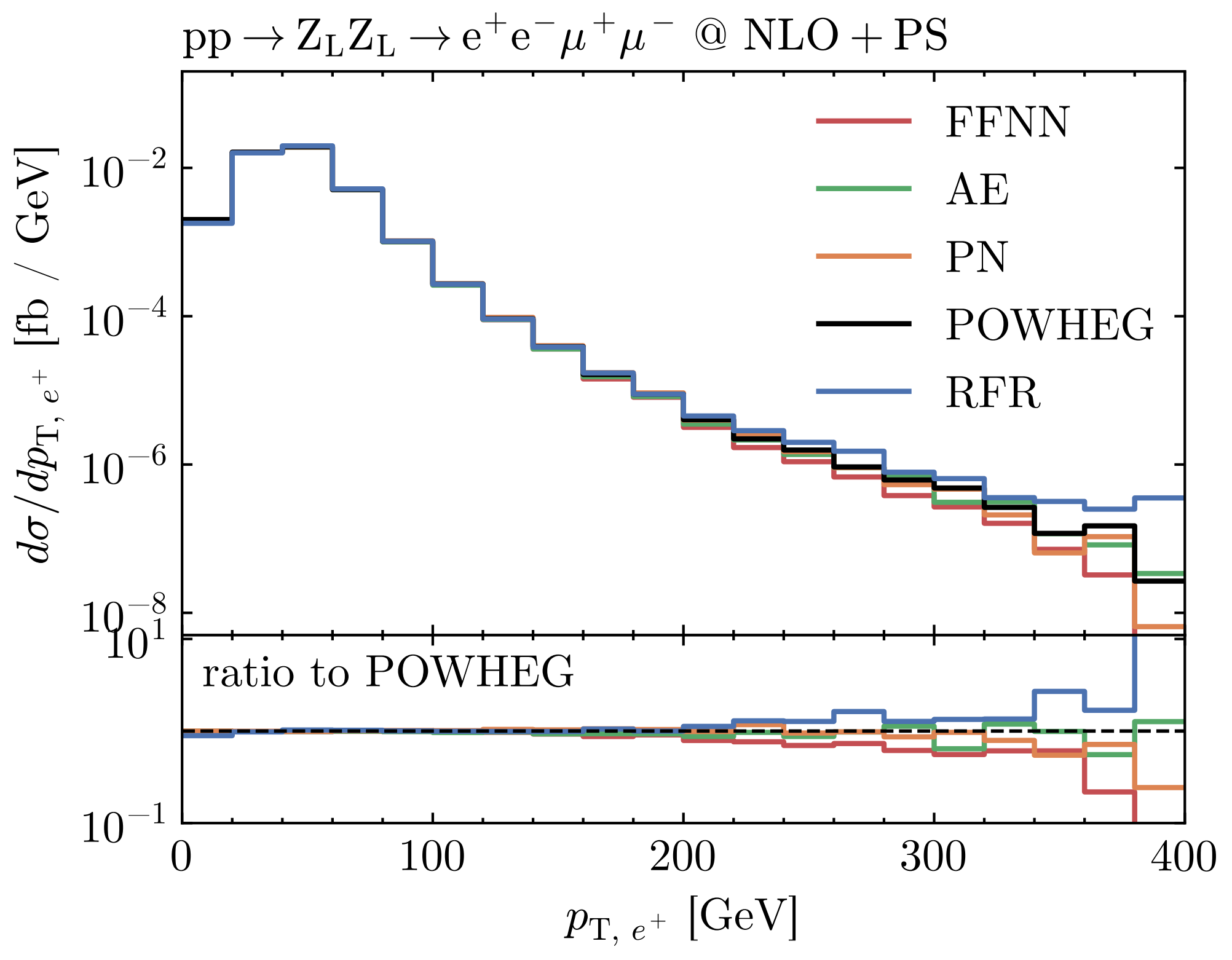
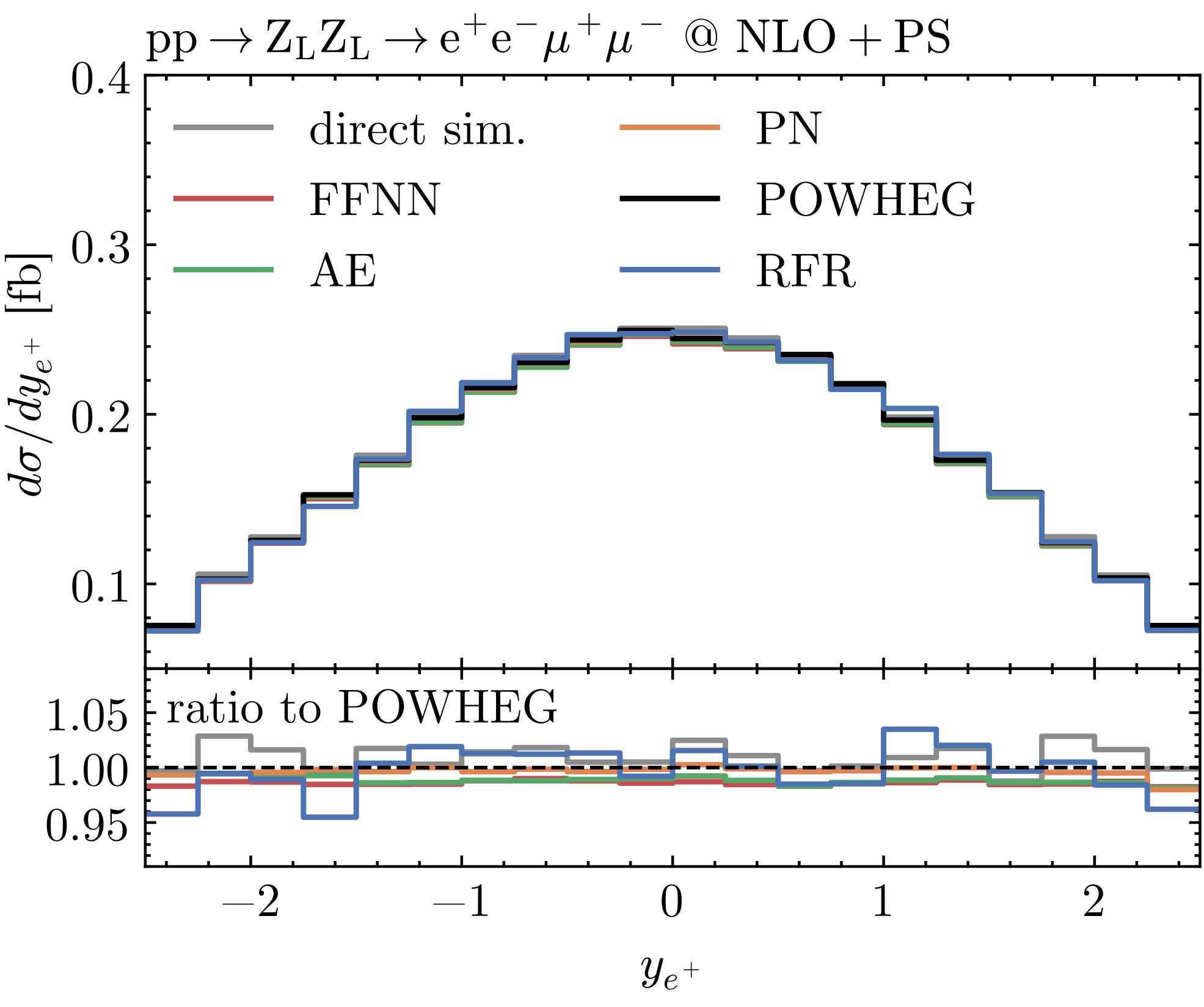
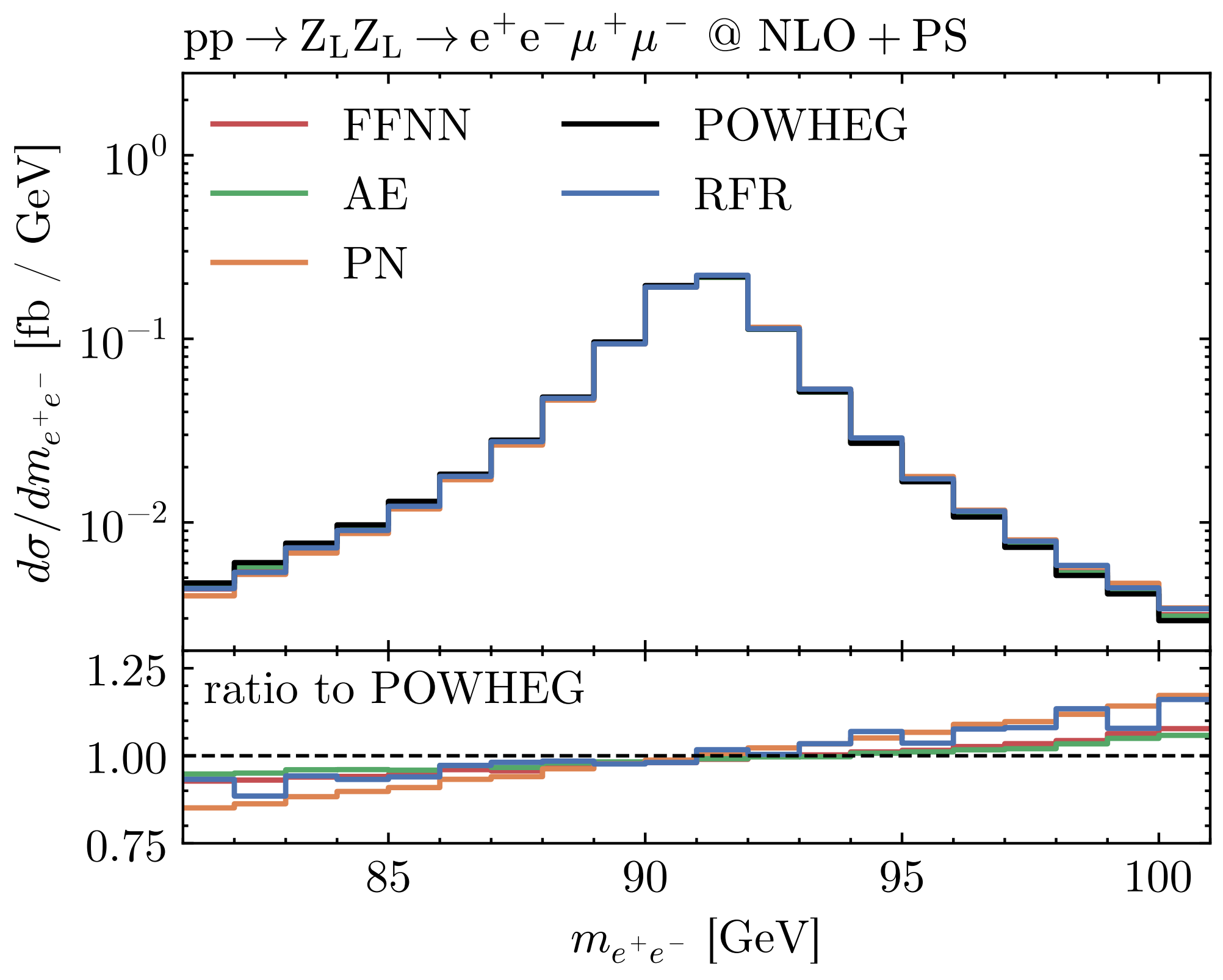
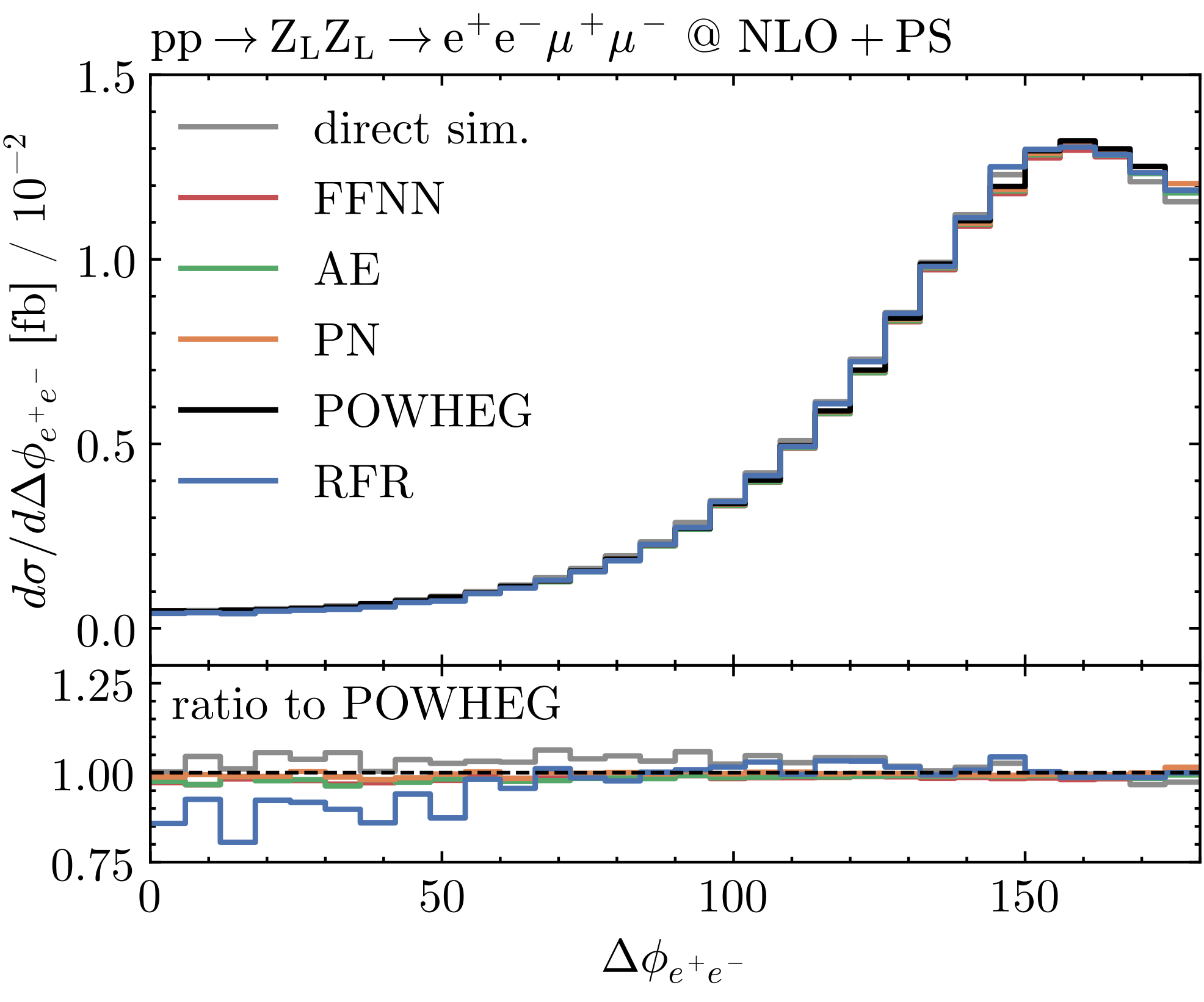

$\mathrm{pp} \to \mathrm{Z_L Z_L} \to \mathrm{e^+ e^- \mu^+ \mu^-}$ @ NLO + PS
FFNN
AE
PN
POWHEG
RFR
$d\sigma / d\cos\theta^*_{e^+}$ [fb]
0.0
0.2
0.4
0.6
0.8
ratio to POWHEG
1.0
1.1
−1.0
−0.5
0.0
0.5
1.0
$\cos\theta^*_{e^+}$

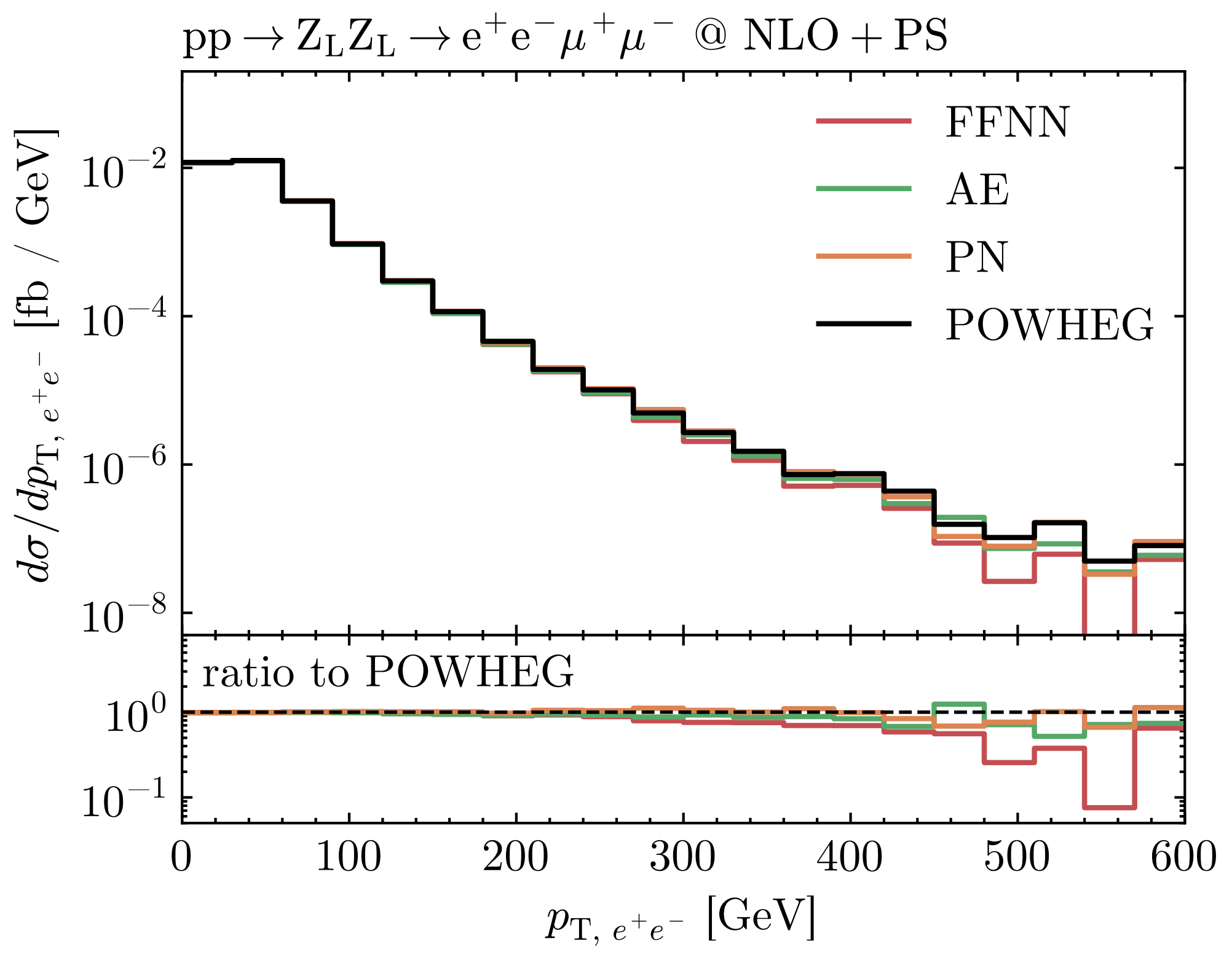

$\mathrm{pp} \to \mathrm{Z_L Z_L} \to \mathrm{e^+ e^- \mu^+ \mu^-}$ @ NLO + PS
FFNN
AE
PN
POWHEG
RFR
$d\sigma / d\cos\theta^*_{e^+}$ [fb]
0.0
0.2
0.4
0.6
0.8
ratio to POWHEG
1.0
1.1
−0.5
0.0
0.5
$\cos\theta^*_{e^+}$

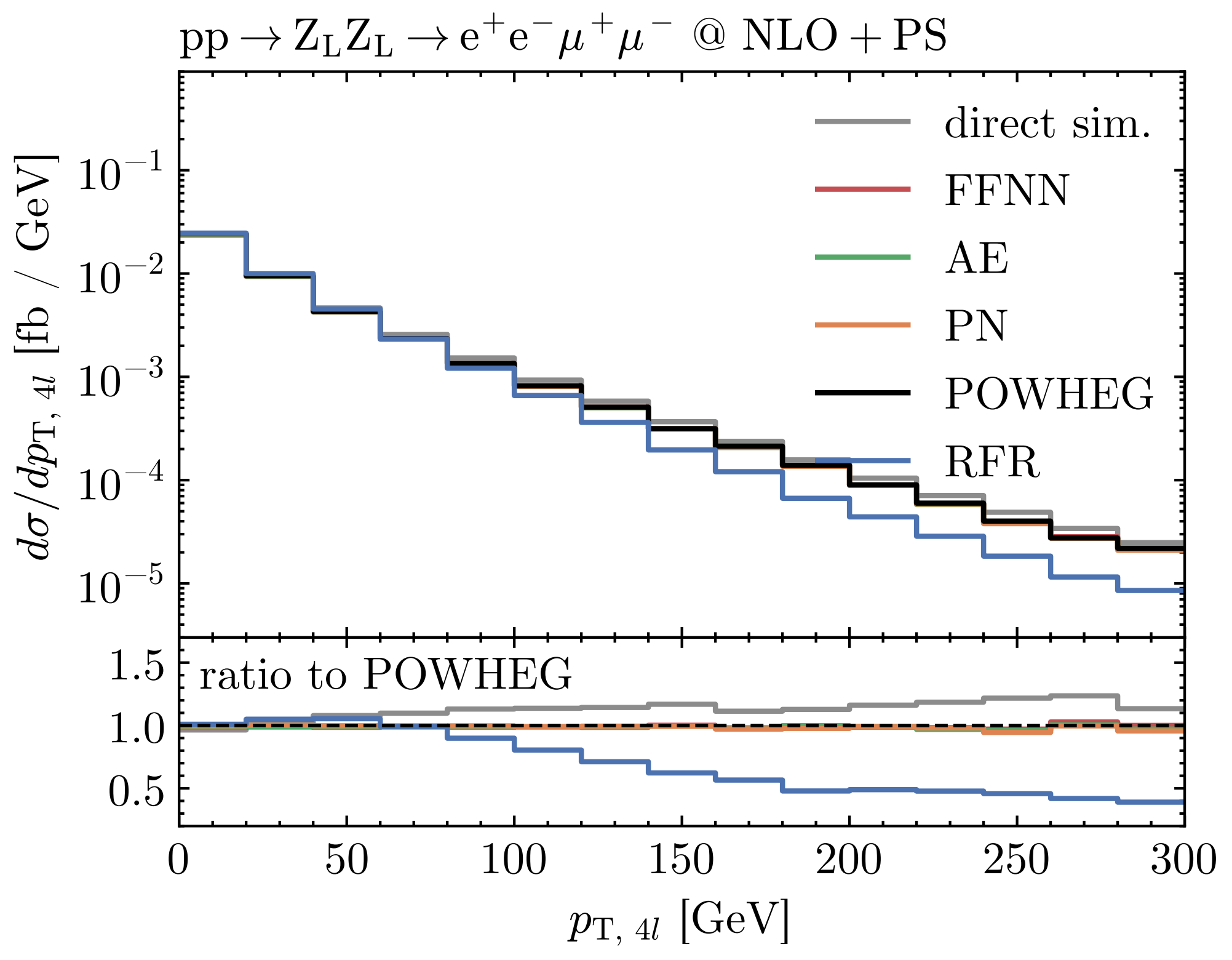

$\mathrm{pp} \to \mathrm{Z_L Z_L} \to \mathrm{e^+ e^- \mu^+ \mu^-}$ @ NLO + PS
FFNN
AE
PN
POWHEG
RFR
$d\sigma/dp_{\mathrm{T},\,e^+}$ [fb / GeV]
10^{-2}
10^{-4}
10^{-6}
10^{-8}
10^{1}
ratio to POWHEG
10^{-1}
0
100
200
300
400
$p_{\mathrm{T},\,e^+}$ [GeV]

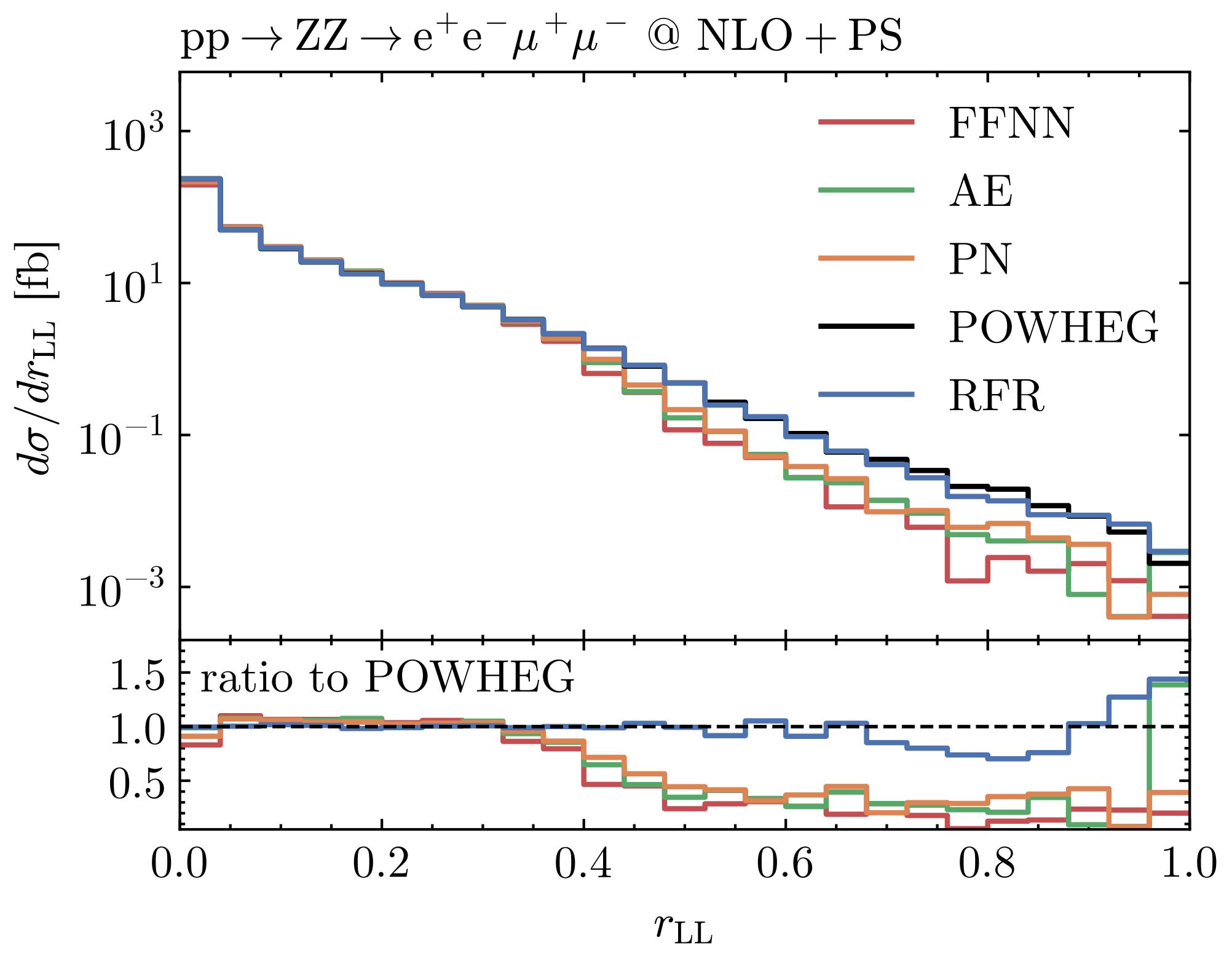

$pp \to Z_L Z_L \to e^+ e^- \mu^+ \mu^-$ @ NLO + PS
direct sim.
FFNN
AE
PN
POWHEG
RFR
$d\sigma/dy_{e^+}$ [fb]
0.4
0.3
0.2
0.1
ratio to POWHEG
1.05
1.00
0.95
−2
−1
0
1
2
y_{e^+}

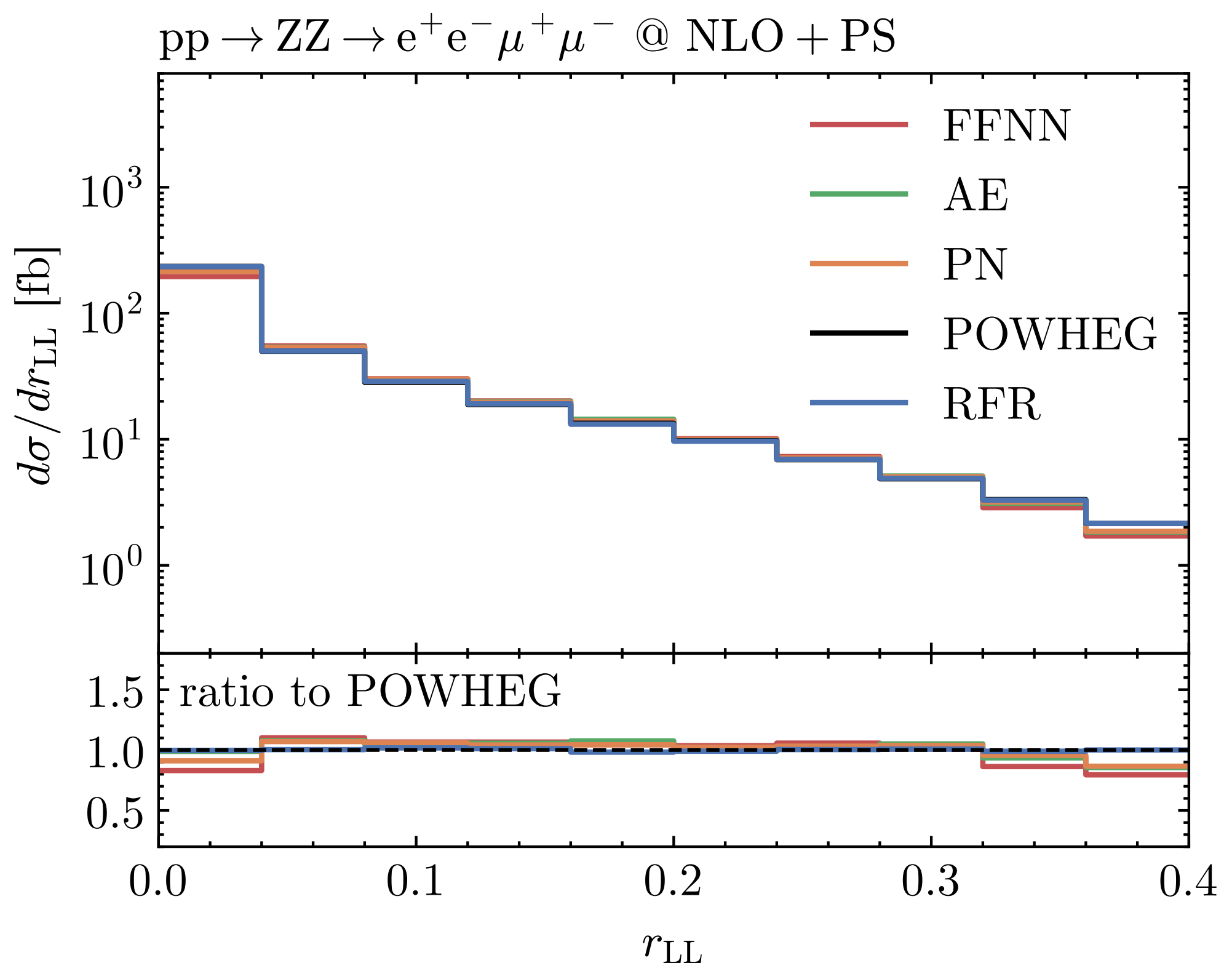

$\mathrm{pp} \to \mathrm{Z_L Z_L} \to \mathrm{e^+ e^- \mu^+ \mu^-}$ @ NLO + PS
FFNN
AE
PN
POWHEG
RFR
$d\sigma/dm_{e^+e^-}$ [fb / GeV]
10^0
10^{-1}
10^{-2}
ratio to POWHEG
1.25
1.00
0.75
85
90
95
100
$m_{e^+e^-}$ [GeV]


$\mathrm{pp} \to \mathrm{Z_L Z_L} \to \mathrm{e^+ e^- \mu^+ \mu^-}$ @ NLO + PS
direct sim.
FFNN
AE
PN
POWHEG
RFR
$d\sigma / d\Delta\phi_{e^+e^-}$ [fb] / 10^{-2}
1.5
1.0
0.5
0.0
ratio to POWHEG
1.25
1.00
0.75
0
50
100
150
$\Delta\phi_{e^+e^-}$


$\mathrm{pp} \to \mathrm{Z_L Z_L} \to \mathrm{e^+ e^- \mu^+ \mu^-}$ @ NLO + PS
FFNN
AE
PN
POWHEG
$d\sigma/dp_{\mathrm{T},\, e^+e^-}$ [fb / GeV]
10^{-2}
10^{-4}
10^{-6}
10^{-8}
ratio to POWHEG
10^{0}
10^{-1}
0
100
200
300
400
500
600
$p_{\mathrm{T},\, e^+e^-}$ [GeV]


$\mathrm{pp} \to \mathrm{Z_L Z_L} \to \mathrm{e^+ e^- \mu^+ \mu^-}$ @ NLO + PS
direct sim.
FFNN
AE
PN
POWHEG
RFR
$d\sigma/dp_{\mathrm{T},\,4l}$ [fb / GeV]
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10^{-5}
ratio to POWHEG
1.5
1.0
0.5
0
50
100
150
200
250
300
$p_{\mathrm{T},\,4l}$ [GeV]


pp → ZZ → e+e−μ+μ− @ NLO + PS
FFNN
AE
PN
POWHEG
RFR
dσ/dr_LL [fb]
10^3
10^1
10^−1
10^−3
ratio to POWHEG
1.5
1.0
0.5
0.0
0.2
0.4
0.6
0.8
1.0
r_LL


pp → ZZ → e+e−μ+μ− @ NLO + PS
FFNN
AE
PN
POWHEG
RFR
dσ/dr_LL [fb]
10^3
10^2
10^1
10^0
ratio to POWHEG
1.5
1.0
0.5
0.0
0.1
0.2
0.3
0.4
r_LL